

JECRC University, Jaipur
Department of Physics
B. Tech Ist Year – Ist Semester July-Dec 2025-26
Applied Physics (DPH001B)
Assignment-III

A. Very Short Answer Questions (10)

1. Name the charge carriers which are responsible for flow of currents in conductors and intrinsic semiconductors? (2)
2. Define the term mobility. Does electron and hole have same mobility? (2)
3. Draw the symbol of PN junction diode, Zener Diode, Varactor Diode, LED and Photodiode. (2)
4. What is the use of Varactor diode in circuit? Also mention how Varactor diode is connected in forward bias or reverse bias condition? (2)
5. What is a photodiode how it is different from LED? (2)

B. Answer the following briefly (21)

1. Find the resistivity of an intrinsic semiconductor with carrier concentration $2.5 \times 10^{19}/\text{m}^3$. The electron and hole mobilities are $0.40 \text{ m}^2/\text{V-s}$ and $0.20 \text{ m}^2/\text{V-s}$ respectively. (7)
2. What is a Zener diode? Explain the breakdown mechanism in Zener diode. How Zener breakdown different from avalanche breakdown. (7)
3. For Intrinsic semiconductors if $m_e = m_h$ prove that the Fermi energy level lies exactly in the middle of highest energy E_v of valance band and lowest energy E_c of the conduction band. (7)

C. Answer the followings explicitly. (33)

1. Derive the expression for electron concentration and hole concentration starting from expression of density of states and Fermi function. Also find the total carrier concentration for intrinsic semiconductor. (11)
2. Derive the expression for conductivity of intrinsic and extrinsic semiconductors. Explain the effect of concentration of minority charge carrier due to addition of impurity in intrinsic semiconductors. (11)
3. Draw a circuit diagram to draw the volt Ampere (V-I) characteristic of PN junction diode. Also draw probable V-I characteristics for PN junction diode and explain different knee voltage and breakdown voltage using the diagram. (11)